

Vol 7 Chapter 5 — Electromagnetic Waves

Keeper (author pass — deep math/physics revision)

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Chapter 5 — Electromagnetic Waves

Level 1 — one sentence

Electromagnetic waves — radio, microwave, infrared, visible light, UV, X-ray, gamma — are propagating wave-packet solutions of the source-free Maxwell equations with speed $c = 1/\sqrt{\mu_0\epsilon_0}$, and in BST they are substrate photon K-type wave packets traveling at the substrate's natural massless-K-type group velocity (the speed of light).

Level 2 — graduate-physicist precision

5.1 The EM wave equation

In vacuum ($\rho = 0, \vec{J} = 0$): taking $\nabla \times$ of Faraday and using Ampère-Maxwell (Chapter 2 Section 2.8):

$$\nabla^2 \vec{E} - \frac{1}{c^2} \partial_t^2 \vec{E} = 0$$

$$\nabla^2 \vec{B} - \frac{1}{c^2} \partial_t^2 \vec{B} = 0$$

with $c^2 = 1/(\mu_0\epsilon_0)$. Both fields satisfy the wave equation; both propagate at speed c .

Plane wave solution: $\vec{E} = \vec{E}_0 \cos(\vec{k} \cdot \vec{r} - \omega t)$ with dispersion $\omega = ck$.

For Maxwell's equations to be satisfied: $\vec{k} \cdot \vec{E}_0 = 0$ (transverse), $\vec{B} = \hat{k} \times \vec{E}/c$ (perpendicular to $\vec{E}$ and $\vec{k}$, with $|\vec{B}| = |\vec{E}|/c$).

5.2 Substrate-mechanism reading: photon K-type wave packets

In BST: EM waves are substrate photon K-type wave packets. The photon is a Pin(2) integer-weight massless K-type (Volume 5 Chapter 3, Vol 7 Chapter 1). Massless = zero substrate Casimir on this K-type sector = lightlike dispersion $\omega = ck$.

Two transverse polarizations correspond to Pin(2) weights ± 1 (helicity). Longitudinal polarization is gauge-equivalent to zero (no physical content).

5.3 The electromagnetic spectrum

Region	Wavelength	Frequency	Typical sources
Radio	> 10 cm	< 3 GHz	Antennas, AM/FM
Microwave	1 mm – 10 cm	3 – 300 GHz	Magnetrons, radar
Infrared	700 nm – 1 mm	0.3 – 430 THz	Thermal emission

Region	Wavelength	Frequency	Typical sources
Visible	380 – 700 nm	430 – 790 THz	LEDs, sun
UV	10 – 380 nm	0.79 – 30 PHz	Sun, UV lamps
X-ray	0.01 – 10 nm	30 PHz – 30 EHz	Bremsstrahlung
Gamma	< 0.01 nm	> 30 EHz	Nuclear decay

All have the same speed $c \approx 3 \times 10^8$ m/s in vacuum; differ only in frequency / wavelength.

Substrate reading: the entire EM spectrum is the substrate's photon K-type wave packet spectrum, from very-low-frequency (large wavelength, low energy per photon) to very-high-frequency (small wavelength, high energy per photon). Same K-type, different excitation level.

5.4 Energy and momentum

EM energy density: $u = (\epsilon_0/2)E^2 + (1/(2\mu_0))B^2 = \epsilon_0 E^2$ (for waves where E and B contribute equally).

EM momentum density: $\vec{p} = \vec{S}/c^2$ where $\vec{S} = \vec{E} \times \vec{B}/\mu_0$ is the Poynting vector (energy flux).

For a wave with field amplitude E_0 : average intensity $\langle S \rangle = (1/2)c\epsilon_0 E_0^2$.

Photon energy: $E_\gamma = \hbar\omega$. Photon momentum: $p_\gamma = \hbar k = E_\gamma/c$. (Massless particle has $E = pc$.)

5.5 Polarization

A plane wave has two independent transverse polarizations. Common bases: - Linear: $\hat{x}$ and $\hat{y}$ (for wave traveling along $\hat{z}$) - Circular: right-handed ($\hat{R} = (\hat{x} + i\hat{y})/\sqrt{2}$) and left-handed ($\hat{L}$) - Elliptical: superposition of linear with phase difference

Polarized waves carry angular momentum: each circular photon carries $\pm\hbar$ along $\hat{k}$.

Substrate reading: linear polarization = substrate K-type with definite SO(2) phase axis; circular = substrate K-type with definite Pin(2) weight ± 1 .

5.6 Reflection and refraction at an interface

Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$, with $n = c/v_{\text{medium}}$ the refractive index.

Fresnel equations give the reflection and transmission amplitudes for waves at an interface; depend on polarization (s and p — perpendicular and parallel to plane of incidence).

Brewster's angle $\theta_B = \arctan(n_2/n_1)$: at this angle, p-polarized light is fully transmitted (no reflection). Used in polarizers and lasers.

5.7 Waves in dielectric media

In a linear dielectric with refractive index n : - Speed: $v = c/n$ - Wavelength: $\lambda = \lambda_0/n$ - Frequency: unchanged (same source) - $\vec{E}, \vec{B}$ amplitudes change by factor $1/\sqrt{\epsilon_r}$

Dispersion: n depends on $\omega \rightarrow$ different frequencies travel at different speeds $\rightarrow$ prism splits light, optical fiber pulse broadening.

5.8 Worked example: visible light wavelength

Yellow-green light (peak of human eye sensitivity): $\lambda \approx 550$ nm, $\nu \approx 5.5 \times 10^{14}$ Hz.

Photon energy: $E_\gamma = h\nu = (6.63 \times 10^{-34})(5.5 \times 10^{14}) \approx 3.6 \times 10^{-19}$ J = 2.25 eV.

For 1 W of yellow-green light (sun-equivalent for $\sim$ few cm^2): photon flux = $1/3.6 \times 10^{-19} \approx 2.8 \times 10^{18}$ photons/sec.

Substrate reading: every second, about 3×10^{18} photon K-types are emitted at this frequency for a 1-W source. The substrate emits these at the Koons-tick rate from the substrate K-type configuration of the emitter atoms.

5.9 The constancy of c

Einstein's 1905 postulate: the speed of light is the same in all inertial frames. This is the foundation of special relativity. EM waves don't have a "medium" — they're pure field excitations that propagate at c regardless of source or observer motion.

Substrate reading: c is the substrate's natural massless-K-type group velocity. The substrate's K-type framework is intrinsically Lorentz-covariant via the $SO_0(4, 2) \subset SO_0(5, 2)$ subgroup (Volume 7 Chapter 7). The constancy of c is the substrate's symmetry, not a separate postulate.

5.10 K-audit anchors

- Chapter 2: Maxwell from substrate $U(1)$ (foundation)
- Volume 5 Chapter 3: photon = $Pin(2)$ integer-weight K-type
- Volume 7 Chapter 7: relativistic EM and $SO(4, 2) \subset SO(5, 2)$

Level 3 — 5th-grader accessibility

EM waves are how electricity and magnetism *move*. Radio, microwave, light, X-ray, gamma — all the same thing at different frequencies. All travel at speed $c \approx 3 \times 10^8$ m/s in vacuum. Wavelength $\times$ frequency = c . Photon energy = $h\nu$ (Planck): radio photons have very little energy, gamma photons have lots. Wave behavior at interfaces: Snell's law for refraction, Fresnel for reflection, Brewster's angle for polarized transmission. In BST, EM waves are substrate photon K-type wave packets — same K-type for the entire spectrum, just different excitation levels. The speed c is the substrate's natural massless-K-type group velocity; it's the same everywhere because it's a substrate property, not a medium property.

What comes next

Chapter 6 develops EM radiation — how accelerated charges emit electromagnetic waves.

Where to look this up

- EM waves: Griffiths Ch 9; Jackson Ch 7-8
- Polarization: Born and Wolf, *Principles of Optics*
- Fresnel equations: Jackson Ch 7
- BST anchors: Chapter 2 Maxwell; Volume 5 Chapter 3 photon K-type